

Paper-2
NCL
(Radiographer)

1. Which part of body is X-rayed in "Frog position"?
 - (a) Pelvis
 - (b) Hip joint
 - (c) Knee joint
 - (d) Elbow joint
2. Which of the following will help in reducing radiation exposure to patients?
 - High KV, low (Milli ampere-seconds) mAs
 - Collimation
 - Filtration
 - (a) 1 and 2 only
 - (b) 2 and 3 only
 - (c) 1 and 3 only
 - (d) 1, 2 and 3
3. The unit of dose equivalent used to express radiation dose in living tissue:-
 - (a) Roentgen
 - (b) Rem
 - (c) Rad
 - (d) Erg
4. TLD badge should be worn at:-
 - (a) Outside the lead apron stomach position
 - (b) Inside the lead apron chest level
 - (c) The back side at the shoulder position
 - (d) Outside the lead apron collar position
5. The common bile duct opens into the:-
 - (a) Medial wall of the first part of duodenum
 - (b) Medial wall of the second part of duodenum
 - (c) Lateral wall of the second part of duodenum
 - (d) Medial wall of the third part of duodenum
6. The stomach is located primarily within the :-
 - (a) Right lower quadrant
 - (b) Right upper quadrant
 - (c) Left upper quadrant
 - (d) Left lower quadrant
7. Normally a patient is asked to inhale deeply when having the lungs x-rayed. Which of the following is the BEST reason for that?
 - (a) More uniform density
 - (b) Increased contrast
 - (c) Greater magnification
 - (d) Greater area of lung structures shown
8. Compression is used during excretory urography in order to accomplish which of the following?
 - (a) Aiding in the excretion process
 - (b) Immobilizing the patient
 - (c) Preventing the patient from breathing during the exposure
 - (d) Retaining the contrast medium in the collecting system
9. Which of the following is a radiographic study of the urinary bladder that involves the direct injection of a contrast media through a catheter?
 - (a) Intravenous pyelogram
 - (b) Intravenous urogram
 - (c) Retrograde pyelogram
 - (d) Cystogram
10. The frequencies of Ultrasound used in medical imaging are in the range of:
 - (a) 1-20 Hz
 - (b) 1-20 MHz
 - (c) 10-20 MHz
 - (d) 10-20 KHz
11. What form of energy is used in MRI?
 - (a) Radio waves
 - (b) Sound waves
 - (c) X-rays
 - (d) **Ultraviolet**

12. The abdominal aorta bifurcates into Left and Right:
(a) Common iliac arteries
(b) Popliteal arteries
(c) Femoral arteries
(d) Tibial arteries
13. Which of the following radiological procedures outlines the Spinal cord?
(a) KUB Radiography
(b) Thermography
(c) Myelography
(d) Xeroradiography
14. Unexposed X-ray film is comprised of a transparent polyester base coated with an emulsion containing radiation-sensitive particle known as:
(a) Silver bromide (b) Silver iodide
(c) Both A and B (d) Neither A or B
15. Swimmer's view is used for taking X-Rays of:
(a) Lumbo-sacral spine
(b) Cervico-thoracic junction
(c) Humerus
(d) Coccyx
16. Waters' view is used for taking X-Rays of:
(a) Base of skull (b) Mastoid bones
(c) Orbits (d) Paranasal sinuses
17. Apicogram is used for taking X-Rays of:
(a) Base of lung (b) Apices of lung
(c) Mediastinum (d) Diaphragm
18. Which procedure uses non-ionizing radiation?
(a) Nuclear medical imaging
(b) X-Ray
(c) CT
(d) MRI
19. To detect gas-fluid level in the abdomen, following X-ray views of abdomen are taken:
(a) AP view
(b) Lateral decubitus view
(c) Erect view
(d) Both B and C
20. The most damaging type of radiation is:
(a) Beta rays (b) X-rays
(c) Alpha rays (d) Gamma rays
21. The following bones participate in the formation of the knee joint:
• Femur
• Tibia
• Patella
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2, and 3
22. A patient is usually required to drink barium sulphate suspension in order to demonstrate which of the following structures?
• Pylorus
• Sigmoid
• Duodenum
(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 3 only
23. The image intensifier's input phosphor is generally composed of:
(a) Cesium iodide
(b) Zinc cadmium sulphide
(c) Gadolinium oxysulphide
(d) Calcium tungstate
24. The following statement(s) is (are) accurate with respect to the differences between the male and female bony pelvis:
• The female pelvic outlet is wider
• The pubic angle is 90 degrees or fewer in the male
• The male pelvis is more shallow
(a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2, and 3
25. With the patient positioned as for a parietoacanthial projection (Waters method), and the central ray directed through the patient's open mouth, which of the following sinus groups is demonstrated through the open mouth?
(a) Frontal (b) Ethmoid
(c) Maxillary (d) Sphenoid
26. In which body position would a patient suffering from orthopnea experience the least discomfort?
(a) Fowler (b) Trendelenburg
(c) Recumbent (d) Erect

27. The four major arteries supplying the brain include the:
- Brachiocephalic artery
 - Common carotid arteries
 - Vertebral arteries
- (a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2, and 3
28. Which of the following is (are) demonstrated in the lateral projection of the thoracic spine?
- Intervertebral joints
 - Apophyseal joints
 - Intervertebral foramina
- (a) 1 only (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only
29. Which of the following conditions is characterized by widening of the intercostal spaces?
- (a) Emphysema (b) Empyema
(c) Atelectasis (d) Pneumonia
30. For AP view of the coccyx, the central ray is directed:
- (a) 25 degrees towards the head
(b) 10 degrees towards the head
(c) 25 degrees towards the feet
(d) 10 degrees towards the feet
31. Typical examples of digital imaging include:
- Magnetic Resonance Imaging (MRI)
 - Computed Tomography (CT)
 - Pluridirectional Tomography
- (a) 1 only (b) 1 and 2 only
(c) 1 and 3 only (d) 1, 2, and 3
32. Of the following groups of technical factors, which will produce the greatest radiographic density?
- (a) 10 mAs, 74 kV, 44-inches SID
(b) 10 mAs, 74 kV, 36-inches SID
(c) 5 mAs, 85 kV, 48-inches SID
(d) 5 mAs, 85 kV, 40-inches SID
33. The energy of ionizing electromagnetic radiations is measured in:
- (a) mA (b) mAs
(c) keV (d) kV
34. A wire mesh is used to test:
- (a) Focal spot size
(b) For screen lag
(c) Film-screen contact
(d) Screen speed
35. Which of the following has an effect on distortion?
- Source-image distance
 - Angulation of the x-ray tube
 - Angulation of the part
- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2, and 3
36. The term used to describe image density in digital imaging is:
- (a) Blackening (b) Brightness
(c) Gray scale (d) Resolution
37. Compared to a low-ratio grid, a higher-ratio grid could have:
- Taller lead strips
 - More distance between the lead strips
 - Thicker lead strips
- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2, and 3
38. Which of the following pathologic conditions would require an increase in exposure factors?
- (a) Pneumoperitoneum
(b) Obstructed bowel
(c) Renal colic
(d) Ascites
39. Grid ratio is defined as the relationship between the height of the lead strip and the:
- (a) Width of the lead strip
(b) Distance between the lead strips
(c) Number of lead strips per inch
(d) Angle of the lead strip
40. Patient dose can be decreased by using:
- High ratio grids
 - High-speed screen and film combination
 - Air-gap technique
- (a) 1 only (b) 1 and 2 only
(c) 3 only (d) 2, and 3

41. In order to better visualize joint space in AP view of knee joint the tube is angled 5 to 7 degrees:
(a) Cranially
(b) Caudally
(c) Towards the medial side
(d) Towards the lateral side
42. MCU is:
(a) Micturating Cysto Ureterography
(b) Modified Cysto Urethrogram
(c) Micturating Cysto Urethrography
(d) Micturating Cysto Urinography
43. The term effective dose refers to:
(a) Whole-body dose
(b) Localized organ dose
(c) Genetic effects
(d) Somatic and genetic effects
44. Examples of primary radiation barriers include:
• X-ray room walls
• Control booth
• Lead aprons
(a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2, and 3
45. When reviewing patient blood chemistry levels, what is considered the normal creatinine range?
(a) 0.6–1.5 mg/100 ml
(b) 4.5–6 mg/100 ml
(c) 8–25 mg/100 ml
(d) Up to 50 mg/100 ml
46. In IVP, the first film after contrast injection is:
(a) Nephrogram (b) Pyelogram
(c) Ureterogram (d) Venogram
47. The number of cervical vertebrae:
(a) 5 (b) 12
(c) 7 (d) 8
48. When X-ray beam falls on the body, they are:
(a) Reflected
(b) Absorbed
(c) Transmitted
(d) All of the above happens
49. The value of CT number (in Hounsfield units) is determined primarily by:
(a) Matrix size (b) Slice thickness
(c) KV (d) Tissue density
50. Anode in rotating anode X-ray tube is made of:
(a) Tungsten-Nickel alloy
(b) Tungsten-Rhenium alloy
(c) Nickel-Tin alloy
(d) Tungsten-Aluminum alloy
51. Which is not true about intensifying screen?
(a) Reduces patient dose
(b) Decreases spatial resolution
(c) Reduces blurring
(d) Increases loading of X-ray tube
52. Ideal temperature of the dark room is:
(a) 10-15 degree C (b) 23-27 degree C
(c) 16-22 degree C (d) 28-32 degree C
53. The most commonly used photo sensitive agent?
(a) Silver Chloride (b) Silver Bromide
(c) Silver Iodide (d) None of the above
54. The specialized radiological examination of lungs and bronchial tree using an opaque contrast medium is called:
(a) Bonchoscopy (b) Bronchography
(c) Cystography (d) Urography
55. To decrease the magnification of the heart in PA view of chest the film focal distance is kept at:
(a) 40 inches (b) 60 inches
(c) 70 inches (d) 36 inches
56. Radiological examination of salivary glands and its ducts by means of contrast medium is termed as:
(a) Ductography (b) Sialography
(c) Sinography (d) Fistulogram
57. Stenver projection view is basically done for:
(a) Base of skull
(b) Petrous portion of temporal bone
(c) Oblique view of mandible
(d) Orbit
58. Which view of the nasal bone shows bony nasal septum best?
(a) Lateral projection
(b) AP projection
(c) Water's projection
(d) None of the above
59. Skyline view is used for:
(a) Talus (b) Scapula
(c) Patella (d) Hip joint

60. What position of the elbow best demonstrates the olecranon process?
 (a) AP view (b) True lateral view
 (c) Axial view (d) Oblique view
61. The patient's chin should be elevated during chest radiography to:
 (a) Avoid superimposition on the apices of lung
 (b) Keep mid sagittal plane parallel
 (c) Reduce patient motion
 (d) Reduce patient dose
62. The 10 days rule for X-ray examination of female patients is applicable to:
 (a) The first 10 days following the cessation of menstruation
 (b) The first 10 days following the onset of menstruation
 (c) The 10 days preceding the menstruation
 (d) 14 days before menstruation
63. In India which of the following is the most significant in the development of radiation protection standard?
 (a) AERB (b) BARC
 (c) NCRP (d) ICRP
64. The radiation warning symbol is a black _____ on a yellow background
 (a) Tri foil (b) Dual foil
 (c) Quad foil (d) Single foil
65. All of the following are potential advantages of using higher KV in radiography except:
 (a) Increased patient exposure
 (b) Reduced X-ray tube heating
 (c) Shorter exposure time
 (d) Decreased image exposure
66. In an adult, the spinal cord usually ends at the level of which vertebra?
 (a) S1 (b) L1
 (c) L5 (d) S3
67. Newton's Inverse Square Law is useful in radiography because it indicates how the radiation intensity is affected by:
 (a) Radioactive decay
 (b) Distance from the source
 (c) The size of the source
 (d) None of the above
68. Fixing solutions used in Radiography are:
 (a) Alkaline (b) Acidic
 (c) Neutral (d) None of the above
69. One rad is equal to the absorption of:
 (a) 1 joules/kg (b) 0.01 Gy
 (c) 100 joules/kg (d) 0.1 Gy
70. The intensity of the X-ray beam that leaves the X-ray tube is not uniform throughout all portions of the beam. This is called:
 (a) Heel effect (b) Thermionic emission
 (c) Stray radiation (d) Tube rating

Answer and Explanation

1. Ans. (b) Hip joint

Exp. The "Frog position" is used to X-ray the hip joint.

2. Ans. (d) 1, 2 and 3

Exp. High kV, low mAs, collimation, and filtration all help in reducing radiation exposure to patients.

3. Ans. (b) Rem

Exp. The unit of dose equivalent used to express radiation dose in living tissue is the rem (Roentgen equivalent man).

4. Ans. (b) Inside the lead apron at the chest level

5. Ans. (b) Medial wall of the second part of duodenum

Exp. The common bile duct opens into the medial wall of the second part of the duodenum.

6. Ans. (c) Left upper quadrant

Exp. The stomach is located primarily within the left upper quadrant of the abdomen.

7. Ans. (d) Greater area of lung structures shown

Exp. Inhaling deeply during a lung X-ray shows a greater area of lung structures.

8. Ans. (d) Retaining the contrast medium in the collecting system

Exp. Compression during excretory urography helps in retaining the contrast medium in the collecting system.

9. Ans. (d) Cystogram

Exp. A cystogram is a radiographic study of the urinary bladder involving the direct injection of contrast media through a catheter.

10. Ans. (b) 1-20 MHz

Exp. The frequencies of ultrasound used in medical imaging are in the range of 1-20 MHz.

11. Ans. (a) Radio waves

Exp. MRI uses radio waves for imaging.

12. Ans. (a) Common iliac arteries

Exp. The abdominal aorta bifurcates into the left and right common iliac arteries.

13. Ans. (c) Myelography

Exp. Myelography is a radiological procedure that outlines the spinal cord.

14. Ans. (c) Both A and B

Exp. Unexposed X-ray film is coated with an emulsion containing silver bromide and silver iodide.

15. Ans. (b) Cervico-thoracic junction

Exp. The swimmer's view is used for taking X-rays of the cervico-thoracic junction.

16. Ans. (d) Paranasal sinuses

Exp. Waters' view is used for taking X-rays of the paranasal sinuses.

17. Ans. (b) Apices of lung

Exp. An apicogram is used for taking X-rays of the apices of the lung.

18. Ans. (d) MRI

Exp. MRI uses non-ionizing radiation.

19. Ans. (d) Both B and C

Exp. To detect gas-fluid levels in the abdomen, lateral decubitus and erect views are taken.

20. Ans. (c) Alpha rays

Exp. Alpha rays are the most damaging type of radiation.

21. Ans. (d) 1, 2, and 3

Exp. The femur, tibia, and patella all participate in the formation of the knee joint.

22. Ans. (b) 1 and 3 only

Exp. Barium sulfate suspension is used to demonstrate the pylorus and duodenum.

23. Ans. (a) Cesium iodide

Exp. The image intensifier's input phosphor is generally composed of cesium iodide.

24. Ans. (b) 1 and 2 only

Exp. The female pelvic outlet is wider, and the pubic angle is 90 degrees or fewer in the male.

25. Ans. (d) Sphenoid

Exp. With the patient positioned for a parietoacanthial projection and the central ray directed through the open mouth, the sphenoid sinus is demonstrated.

26. Ans. (d) Erect

Exp. A patient suffering from orthopnea experiences the least discomfort in the erect position.

27. Ans. (c) 2 and 3 only

Exp. The common carotid arteries and vertebral arteries are among the major arteries supplying the brain.

28. Ans. (c) 1 and 3 only

Exp. The intervertebral joints and intervertebral foramina are demonstrated in the lateral projection of the thoracic spine.

29. Ans. (a) Emphysema

Exp. Emphysema is characterized by the widening of the intercostal spaces.

30. Ans. (d) 10 degrees towards the feet

Exp. For the AP view of the coccyx, the central ray is directed 10 degrees towards the feet.

31. Ans. (b) 1 and 2 only

Exp. Typical examples of digital imaging include MRI and CT.

32. Ans. (b) 10 mAs, 74 kV, 36-inches SID

Exp. This group of technical factors will produce the greatest radiographic density.

33. Ans. (c) keV

Exp. The energy of ionizing electromagnetic radiations is measured in keV (kilo-electron volts).

34. Ans. (c) Film-screen contact

Exp. A wire mesh is used to test film-screen contact.

35. Ans. (d) 1, 2, and 3

Exp. Source-image distance, angulation of the x-ray tube, and angulation of the part all have an effect on distortion.

36. Ans. (b) Brightness

Exp. The term used to describe image density in digital imaging is brightness.

37. Ans. (a) 1 only

Exp. A higher-ratio grid could have taller lead strip compared to a low-ratio grid.

38. Ans. (d) Ascites

Exp. Ascites would require an increase in exposure factor

39. Ans. (b) Distance between the lead strips

Exp. Grid ratio is defined as the relationship between the height of the lead strip and the distance between the lead strips.

40. Ans. (d) 2 and 3**41. Ans. (b) Caudally**

Exp. To better visualize the joint space in the AP view of the knee joint, the tube is angled 5 to 7 degrees caudally

42. Ans. (c) Micturating Cysto Urethrography

Exp. MCU stands for Micturating Cysto Urethrography

43. Ans. (a) Whole-body dose

44. **Ans. (a) 1 only**
Exp. Primary radiation barriers include X-ray room walls.
45. **Ans. (a) 0.6–1.5 mg/100 ml**
Exp. The normal creatinine range is 0.6–1.5 mg/100 ml.
46. **Ans. (a) Nephrogram**
Exp. In IVP, the first film after contrast injection is a nephrogram.
47. **Ans. (c) 7**
Exp. The number of cervical vertebrae is 7.
48. **Ans. (d) All of the above happens**
Exp. When the X-ray beam falls on the body, it can be reflected, absorbed, or transmitted.
49. **Ans. (d) Tissue density**
Exp. The value of the CT number (in Hounsfield units) is determined primarily by tissue density.
50. **Ans. (b) Tungsten-Rhenium alloy**
Exp. The anode in a rotating anode X-ray tube is made of a tungsten-rhenium alloy.
51. **Ans. (c) Reduces blurring**
Exp. It is not true that the intensifying screen reduces blurring.
52. **Ans. (c) 16–22 degree C**
Exp. The ideal temperature of the dark room is 16–22 degrees Celsius.
53. **Ans. (b) Silver Bromide**
Exp. The most commonly used photosensitive agent is silver bromide.
54. **Ans. (b) Bronchography**
Exp. The specialized radiological examination of lungs and bronchial tree using an opaque contrast medium is called bronchography.
55. **Ans. (c) 70 inches**
Exp. To decrease the magnification of the heart in the PA view of the chest, the film focal distance is kept at 70 inches.
56. **Ans. (b) Sialography**
Exp. The radiological examination of salivary glands and their ducts by means of a contrast medium is termed sialography.
57. **Ans. (b) Petrous portion of the temporal bone**
Exp. Stenver projection view is basically done for the petrous portion of the temporal bone.
58. **Ans. (c) Water's projection**
Exp. Water's projection best shows the bony nasal septum.
59. **Ans. (c) Patella**
Exp. Skyline view is used for the patella.
60. **Ans. (b) True lateral view**
Exp. The true lateral view of the elbow best demonstrates the olecranon process.
61. **Ans. (a) Avoid superimposition on the apices of lung**
Exp. The patient's chin should be elevated during chest radiography to avoid superimposition on the apices of the lung.
62. **Ans. (b) The first 10 days following the onset of menstruation**
Exp. The 10 days rule for X-ray examination of female patients is applicable to the first 10 days following the onset of menstruation.
63. **Ans. (a) AERB**
Exp. In India, the Atomic Energy Regulatory Board (AERB) is most significant in the development of radiation protection standards.
64. **Ans. (a) Tri foil**
Exp. The radiation warning symbol is a black tri-foil on a yellow background.
65. **Ans. (d) Decrease image exposure**
66. **Ans. (b) L1**
Exp. In an adult, the spinal cord usually ends at the level of the L1 vertebra.
67. **Ans. (b) Distance from the source**
Exp. Newton's Inverse Square Law indicates how the radiation intensity is affected by the distance from the source.
68. **Ans. (b) Acidic**
Exp. Fixing solutions used in radiography are acidic.
69. **Ans. (b) 0.01 Gy**
Exp. One rad is equal to the absorption of 0.01 Gy.
70. **Ans. (a) Heel effect**
Exp. The intensity of the X-ray beam that leaves the X-ray tube is not uniform throughout all portions of the beam, known as the heel effect.